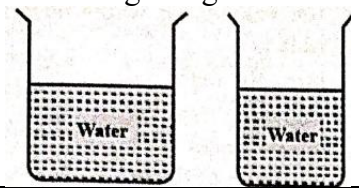
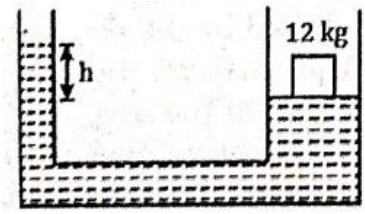
Question	For the figures given below, the correct observation is: 
Type	multiple choice
Option	Pressure at the bottom of the 1st tank is greater than at the bottom of the 2nd tank.
Option	Pressure at the bottom of the 1st tank is lesser than at the bottom of 2nd tank.
Option	Pressure depends upon the shape of the container.
Option	Pressure at the bottom of 1st tank and 2nd tank are the same.
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	A diver is 10 m below the surface of water. The approximate pressure experienced by the diver is-
Type	multiple choice
Option	10^5 Pa
Option	2×10^5 Pa
Option	3×10^5 Pa
Option	4×10^5 Pa
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	If pressure at half the depth of a lake is equal to 2/3rd the pressure at the bottom of the lake, then the depth of the lake is:
Type	multiple choice
Option	10m
Option	20m
Option	60m
Option	30m
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	The area of cross-section of the wider tube shown in the figure is 800 cm². If mass of 12kg is placed on the massless piston, then the difference in heights h of the levels of water in the two tubes will be: 
Type	multiple choice
Option	10 cm
Option	6 cm
Option	15 cm
Option	2 cm
Answer	3

Solution	
Positive Marks	4
Negative Marks	1
Question	If two iron balls when fully immersed in water experience thrust force in the ratio of the 1:2 then of the masses of the balls will be:
Type	multiple choice
Option	$1:1$
Option	$1:2$
Option	$2:1$
Option	$1:4$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	If two pipes of metals when immersed in a liquid have equal upthrust on them, then
Type	multiple choice
Option	Both pieces must have equal weights
Option	Both pieces must have equal densities
Option	Both pieces must have equal volumes
Option	Both are floating in the same depth
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	A block of ice floats on a liquid of density $1.2\ g/m^3$ in a beaker. The level of liquid when ice completely melts-
Type	multiple choice
Option	Remains same
Option	Rises
Option	Lowers
Option	(1), (2) or (3)
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Two bodies are in equilibrium when suspends in water from the arms of a balance. The mass of one body is 36 g and its density is $9g/cm^3$. If the mass of the other is 48 g, its density in g/cm^3 will be
Type	multiple choice
Option	$\frac{4}{3}$
Option	$\frac{3}{2}$
Option	3
Option	5
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Pascal's law states that pressure at a point is equal in all directions:

Type	multiple choice
Option	In a turbulent flow
Option	In a fluid in motion
Option	In a fluid at rest
Option	In a laminar flow
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	The hydraulic lift is based on which of the following?
Type	multiple choice
Option	Bernoulli's principle
Option	Pascal's law
Option	Archimedes' principle
Option	Boyle's law
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	An ideal fluid (laminar flow) through a pipe of non-uniform diameter. The maximum and minimum diameters of the pipes are 6.4 cm and 4.8 cm respectively. The ratio of the minimum and the maximum velocities of fluid in this pipe is.
Type	multiple choice
Option	$\frac{9}{16}$
Option	$\frac{3}{4}$
Option	3
Option	4
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	The pressure of water in a water pipe when tap is opened and closed are respectively $3\times10^5\,\text{Nm}^{-2}$ and $3.5\times10^5\,\text{Nm}^{-2}$.with open tap, the velocity of water flowing is:
Type	multiple choice
Option	$10\ \text{ms}^{-1}$
Option	$5\ \text{ms}^{-1}$
Option	$20\ \text{ms}^{-1}$
Option	$15\ \text{ms}^{-1}$
Answer	1
Solution	

Positive Marks	4
Negative Marks	1
Question	In horizontal pipe line the pressure falls by $8\,\mathrm{Nm}^{-2}$ between two points separated by a distance of 1 km. Density of oil is $800\,\mathrm{kg\,m}^{3}$. Change in kinetic energy per kg of the oil flowing in the tubes is:
Type	multiple choice
Option	$10^{3}\,\mathrm{J/kg}$
Option	$10^{2}\,\mathrm{J/kg}$
Option	$10^{-2}\,\mathrm{J/kg}$
Option	$10^{-3}\,\mathrm{J/kg}$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	A tank has an orifice near its bottom. The volume of the liquid flowing per second out of the orifice does not depends upon:
Type	multiple choice
Option	The area of the orifice.
Option	The height of the liquid level above the orifice.
Option	The density of the liquid.
Option	Acceleration due to gravity.
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	There is a hole in the bottom of a tank having water. If the total pressure at the bottom is 3 atm $\left(1\text{ atm} = 10^{5}\,\mathrm{N/m^{2}}\right)$, then the velocity of water flowing from the hole is:
Type	multiple choice
Option	$\sqrt{400}\,\mathrm{m/s}$
Option	$\sqrt{600}\,\mathrm{m/s}$
Option	$\sqrt{60}\,\mathrm{m/s}$
Option	None of these
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	The velocity of kerosene oil in a horizontal pipe is 5 m/s. If $g = 10\,\mathrm{m/s^{2}}$, then the velocity head of oil will be:
Type	multiple choice
Option	1.25 m
Option	12.5 m
Option	0.125 m
Option	125 m
Answer	1
Solution	
Positive Marks	4
Negative Marks	1

Question	The value of Reynold's number for which the flow of water through a pipe becomes turbulent is:
Type	multiple choice
Option	500
Option	900
Option	750
Option	6000
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	The velocity of water in a river is
Type	multiple choice
Option	Same everywhere.
Option	More in the middle and less near its banks
Option	Less in the middle and more at banks
Option	Increase from one bank to other.
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Equation of continuity based on:
Type	multiple choice
Option	Conservation of mass
Option	Conservation of energy
Option	Conservation of angular momentum
Option	None of these
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	The weight of an aeroplane flying in the air is balanced by
Type	multiple choice
Option	Vertical component of the thrust created by air currents striking the lower surface of the wings
Option	Force due to reaction of gasses ejected by the revolving propeller
Option	Upthrust of the air which will be equal to the weight of the air having the same volume as the plane
Option	Force due to the pressure difference between the upper and lower surface of the wings created by different air speeds on the surfaces
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Two spheres of the same material, but of radii R and 3R are allowed to fall vertically downwards through a liquid of density σ. The ratio of their terminal velocities is
Type	multiple choice
Option	$1:3$
Option	$1:6$
Option	$1:9$
Option	$1:1$
Answer	3

Solution	
Positive Marks	4
Negative Marks	1
Question	A square plate of 0.1 m side moves parallel to a second plate with a velocity of 0.1 m/s, both plates being immersed in water. If the viscous force is 0.002 N and the coefficient of viscosity is 0.01 poise. Distance between the plates in m is
Type	multiple choice
Option	0.1
Option	0.05
Option	0.005
Option	0.0005
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	The velocity of the surface layer of water in a river of depth 10 m is $5\,ms^{-1}$. The shearing stress between the surface layer and the bottom layer is (Coefficient of viscosity of water, $\eta = 10^{-3}$ SI units)
Type	multiple choice
Option	$0.6\times10^{-3}\ Nm^{-2}$
Option	$0.8\times10^{-3}\ Nm^{-2}$
Option	$0.5\times10^{-3}\ Nm^{-2}$
Option	$10^{-3}\ Nm^{-2}$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Eight drops of a liquid of density p and each of radius a are falling through air with a constant velocity $3.75\,cms^{-1}$. When the eight drops coalesce to form a single drop the terminal velocity of the new drop will be
Type	multiple choice
Option	$15\times10^{-2}m^{-1}$
Option	$2.4\times10^{-2}m^{-1}$
Option	$0.75\times10^{-2}m^{-1}$
Option	$25\times10^{-2}m^{-1}$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Two solid spheres of same metal but of mass M and 8M fall simultaneously on a viscous liquid and their terminal velocities are v and nv, then value of n is
Type	multiple choice
Option	16
Option	8
Option	4
Option	2
Answer	3
Solution	
Positive Marks	4
Negative Marks	1

Question	Spherical ball of radius 'r' is falling in a viscous fluid of viscosity $'\eta'$ with a velocity 'v'. The retarding viscous force acting on the spherical ball is
Type	multiple choice
Option	Inversely proportional to 'r' but directly proportional to velocity 'v'
Option	Directly proportional to both radius 'r' and velocity 'v'
Option	Inversely proportional to both radius 'r' and velocity 'v'
Option	Directly proportional to 'r' but inversely proportional to 'v'
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Two rain drops of same radii 'r', falling with terminal velocity 'v' merge and form a bigger drop of radius R. The terminal velocity of the bigger drop is
Type	multiple choice
Option	$v\frac{R}{r}$
Option	$v\frac{R^2}{r^2}$
Option	v
Option	$2v$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	A small spherical ball falling through a viscous medium of negligible density has terminal velocity v. Another ball of the same mass but of radius twice of the earlier falling through the same viscous medium will have terminal velocity
Type	multiple choice
Option	v
Option	$v/4$
Option	$v/2$
Option	$2v$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	The terminal speed attained by an aluminium sphere of radius 1mm falling through water at 20°C will be close to (Assume laminar flow, specific gravity of $Al = 2.7$ and $\eta_{water} = 8\times10^{-4}$)
Type	multiple choice
Option	$9.2\ m/s$
Option	$6.9\ m/s$
Option	$4.6\ m/s$
Option	$2.3\ m/s$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Two metal spheres are falling through a liquid of density $2\times10^3 kg/m^3$ with the same uniform speed. The material density of sphere 1 and sphere 2 are

	$8\times10^3 kg/m^3$ and $11\times10^3 kg/m^3$ respectively. The ratio of their radii is
Type	multiple choice
Option	$\frac{11}{8}$
Option	$\sqrt{\frac{11}{8}}$
Option	$\frac{3}{2}$
Option	$\sqrt{\frac{3}{2}}$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Shape of meniscus for a liquid of zero angle of contact is-
Type	multiple choice
Option	Plane
Option	Parabolic
Option	hemi-spherical
Option	Cylindrical
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	The force required to separate two glass plates of area $10^{-2} m^2$ with a film of water 0.05 mm thick between them, is (Surface tension of water is $70\times10^{-3} N/m$)
Type	multiple choice
Option	28 N
Option	14 N
Option	50 N
Option	38 N
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	The maximum force, in addition to the weight required to pull a wire of 5.0 cm long from the surface of water at temperature $20^0 C$, is 728 dynes. The surface tension of water is
Type	multiple choice
Option	7.28 N/cm
Option	7.28 dyne/cm
Option	72.8 dyne/cm
Option	7.28×10^3 dyne /cm
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	A false statement is :
Type	multiple choice

Option	Angle of contact $\theta < 90°$, if cohesive force < adhesive force
Option	Angle of contact $\theta > 90°$, if cohesive force > adhesive force
Option	Angle of contact $\theta = 90°$, If cohesive force = adhesive force
Option	If the radius of capillary is reduced to half, if the rise of liquid column becomes four times
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	A liquid drop of diameter D breaks into 27 tiny drops. The resultant change in energy is
Type	multiple choice
Option	$2\pi TD^2$
Option	$4\pi TD^2$
Option	πTD^2
Option	None of these
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	The work done in blowing a soap bubble of radius 0.2 m, given that the surface tension of soap solution is $60\times10^{-3}\,N/m$, is
Type	multiple choice
Option	$24\pi\times10^{-4}\,J$
Option	$8\pi\times10^{-4}\,J$
Option	$96\pi\times10^{-4}\,J$
Option	$192\pi\times10^{-4}\,J$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Two small drops of mercury, each of radius R, coalesce to from a single drop. The ratio of the total surface energies before and after change is
Type	multiple choice
Option	$1:2^{1/3}$
Option	$2^{1/3}:1$
Option	$2:1$
Option	$1:2$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	If the surface tension of water is 0.06 Nm, then capillary rise in a tube of diameter 1 mm is $\left(\theta = 0^0\right)$
Type	multiple choice
Option	1.22 cm
Option	2.4 cm
Option	3.12 cm
Option	3.86 cm
Answer	2

Solution	
Positive Marks	4
Negative Marks	1
Question	Two capillary tubes of same diameter are put vertically one each in two liquids whose relative densities are 0.8 and 0.6 and surface tension are 60 dyne/cm and 50 dyne/cm respectively. Ratio of heights of liquids in the two tubes h_1 / h_2 is
Type	multiple choice
Option	$\frac{10}{9}$
Option	$\frac{3}{10}$
Option	$\frac{10}{3}$
Option	$\frac{9}{10}$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Excess pressure of one soap bubble is four times that of other. Then the ratio of volume of first bubble to another one is
Type	multiple choice
Option	$1:64$
Option	$1:4$
Option	$64:1$
Option	$1:2$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1